

1. (a) $U(8)$ is abelian but not cyclic. So is $\mathbb{R}$ under $+$

(b) 1st choose any nonabelian group such as D_4

The set of rotations in D_4 is an abelian subgroup of D_4

(c) Need any 3 two by two matrices with real entries $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ such that $ad - bc \neq 0$

2. (a) $6 \cdot 5 = 2$

(b) $C(5) = \{1, 3, 5, 7\}$

(c) $H = \{1, 2, 3, 4\}$ $K = \{1, 3, 5, 7\}$

3. $\mathbb{Z}_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$U(10) = \{1, 3, 7, 9\}$. $U(10)$ is not a subgroup of $\mathbb{Z}_{10}$ since the operation in $\mathbb{Z}_{10}$ is addition mod 10 but the operation in $U(10)$ is multiplication mod 10. $U(10)$ is not closed under addition mod 10

4. (a) In $\mathbb{Z}_{40}$ $\langle 4 \rangle = \{0, 4, 8, 12, 16, 20, 24, 28, 32, 36\}$ is a subgroup of order 10.

(b) Elements of order 10 are 4, 12, 28, and 36.

(c) All possible orders in $\mathbb{Z}_{40}$ are 1, 2, 4, 5, 8, 10, 20, and 40.

5. (a) $x^{10} = e$ means $|x|$ divides 10. $x^{21} = e$ means $|x|$ divides 21

The only common divisor of 10 and 21 is 1.

The only element of order 1 is the identity so $x = e$

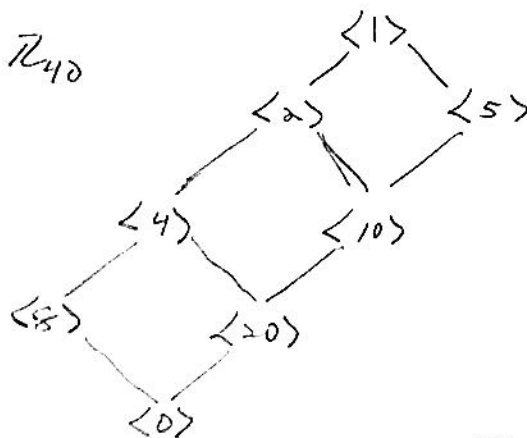
(b) If $|y| = 10$ we know $|G|$ is a multiple of 10.

If $|z| = 21$ we know $|G|$ is a multiple of 21

so $|G|$ must be a common multiple of 10 and 21

$|G|$ is a multiple of 210.

6. Subgroup lattice for $\mathbb{Z}_{40}$



7. $|a| = 48$

$k = 8$

(a) $|a^8| = 6$

(b) $|a^{10}| = \frac{48}{\gcd(48, 10)} = \frac{48}{2} = 24$

$n = 2$

8. (a) False $\gcd(x, y)$ could be 1, 2, 5 or 10.

(b) True. $Z(G)$ = set of elements in G that commute with every element of G . So if $Z(G) = G$, every element of G commutes with every element of G . Thus G is abelian.

9. Let a, b be integers with $\gcd(a, b) = d$. So by the linear combination theorem there are integers x and y such that $ax + by = d$. Dividing through by d we get

$\frac{a}{d}x + \frac{b}{d}y = 1$. Thus 1 can be written as a linear combination of the integers $\frac{a}{d}$ and $\frac{b}{d}$. Since 1 is the smallest positive integer that can be written as a linear combination of $\frac{a}{d}$ and $\frac{b}{d}$, we know $\gcd(\frac{a}{d}, \frac{b}{d}) = 1$.

10. Prove the identity in any group is unique.

Suppose e and e' both satisfy $ex = x = xe$ and $e'x = x = xe'$ for all x in G .

Then letting $x = e'$ we get $e = e \cdot e'$ by (1) and $e \cdot e' = e'$ by (2)

so $e = e'$

11. Given that G is abelian. Define $H = \{x \in G : x^3 = e\}$

H is a subset of G by definition and $e \in H$ since $e^3 = e$.
So H is nonempty.

Let x and y be any two elements of H . This means
 $x^3 = e$ and $y^3 = e$. $(xy)^3 = (xy)(xy)(xy) \stackrel{G \text{ is abelian}}{=} (x \cdot x \cdot x)(y \cdot y \cdot y) = x^3 \cdot y^3$
 $= e \cdot e = e$

Therefore $xy \in H$ so H is closed under the operation.

$$(x^{-1})^3 = (x^3)^{-1} \text{ since } x^3 \cdot (x^{-1})^3 = x \cdot x \cdot x \cdot x^{-1} \cdot x^{-1} \cdot x^{-1} = e = x^3 \cdot (x^3)^{-1}$$

Thus $(x^{-1})^3 = (x^3)^{-1} = e^{-1} = e$. Therefore $x^{-1} \in H$.

By the two step subgroup test H is a subgroup of G .

$$12. \mathcal{U}(25) = \{1, 2, 3, 4, 6, 7, 8, 9, 11, 12, 13, 14, 16, 17, 18, 19, 21, 22, 23, 24\}$$

$|\mathcal{U}(25)| = 20$ so the order of each element must be a
divisor of 20.

To prove $\mathcal{U}(25)$ is cyclic we need to find a generator, an
element with order 20.

Try 2

k	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
$2^k \text{ mod } 25$	2	4	8	16	7	14	3	6	12	24	23	21	17	9	18	11	22	19	13	1

$$\mathcal{U}(25) = \langle 2 \rangle$$

Generators are $2^1, 2^3, 2^7, 2^9, 2^{11}, 2^{12}, 2^{17}$, and 2^{19}

$2, 8, 3, 12, 23, 17, 22$, and 13